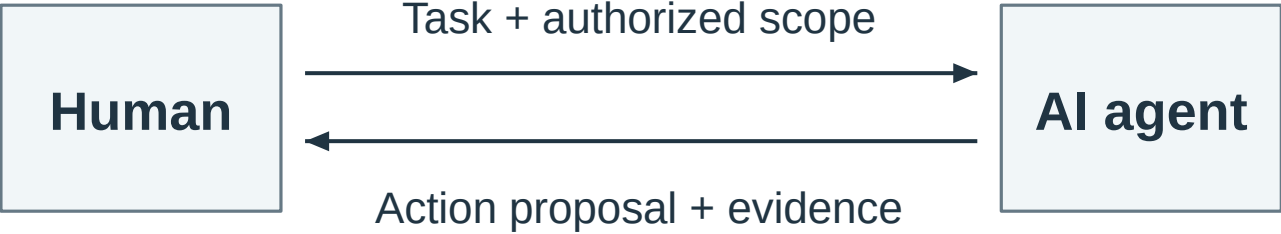


Reliable AI Agents Under Human Control

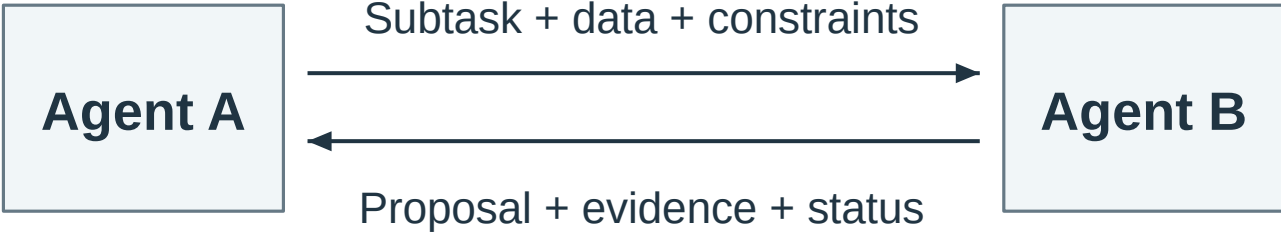
A proposed research agenda for actions, system updates, and delegation

Human-Agent Interaction



- I Check actions against approved requirements
- II Evaluate protection after system updates
- III Enforce current scope and permissions across steps

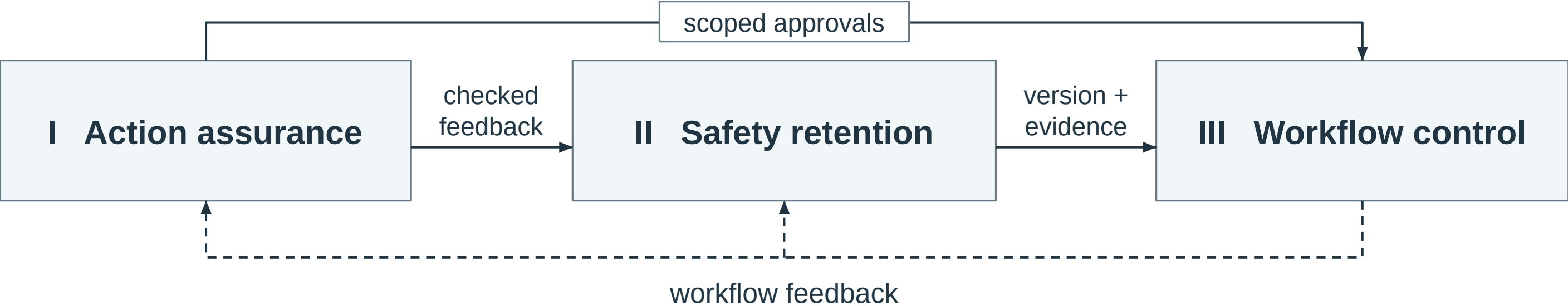
Agent-Agent Collaboration



- I Check actions using independent evidence
- II Assess retention after agent or harness updates
- III Enforce inherited data limits and valid delegation

Shared human-approved requirements Φ

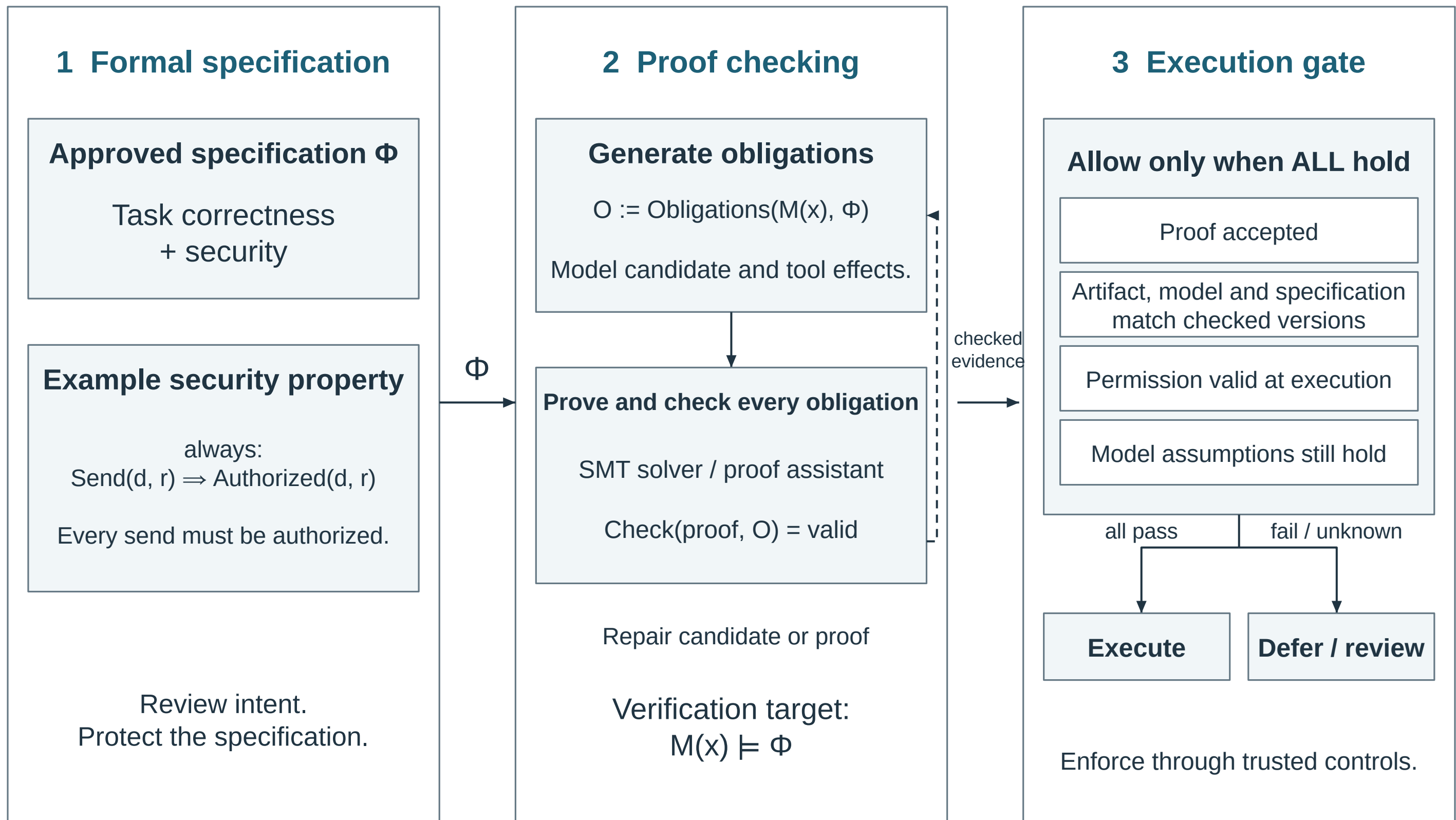
Task criteria · security properties · authorization scope



Formal claims require faithful models, sound checking, and trusted enforcement;
tests provide evidence for evaluated conditions.

I. Formal Oversight Before Action

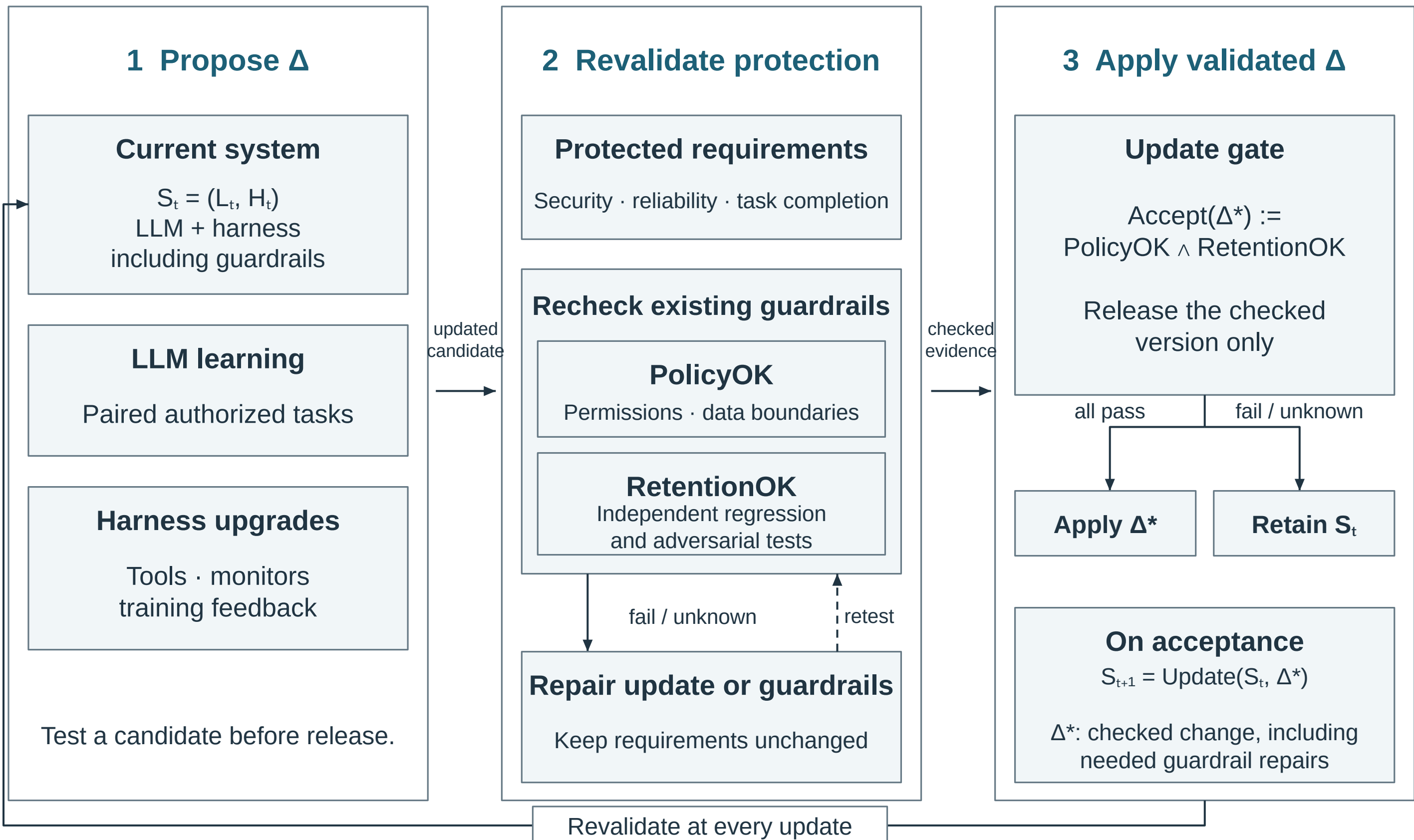
Proposed framework: specify properties, prove obligations, enforce execution



Assurance requires adequate specifications, faithful models, sound proof checking, and trusted enforcement.

II. Preserving Safety Through Continual Updates

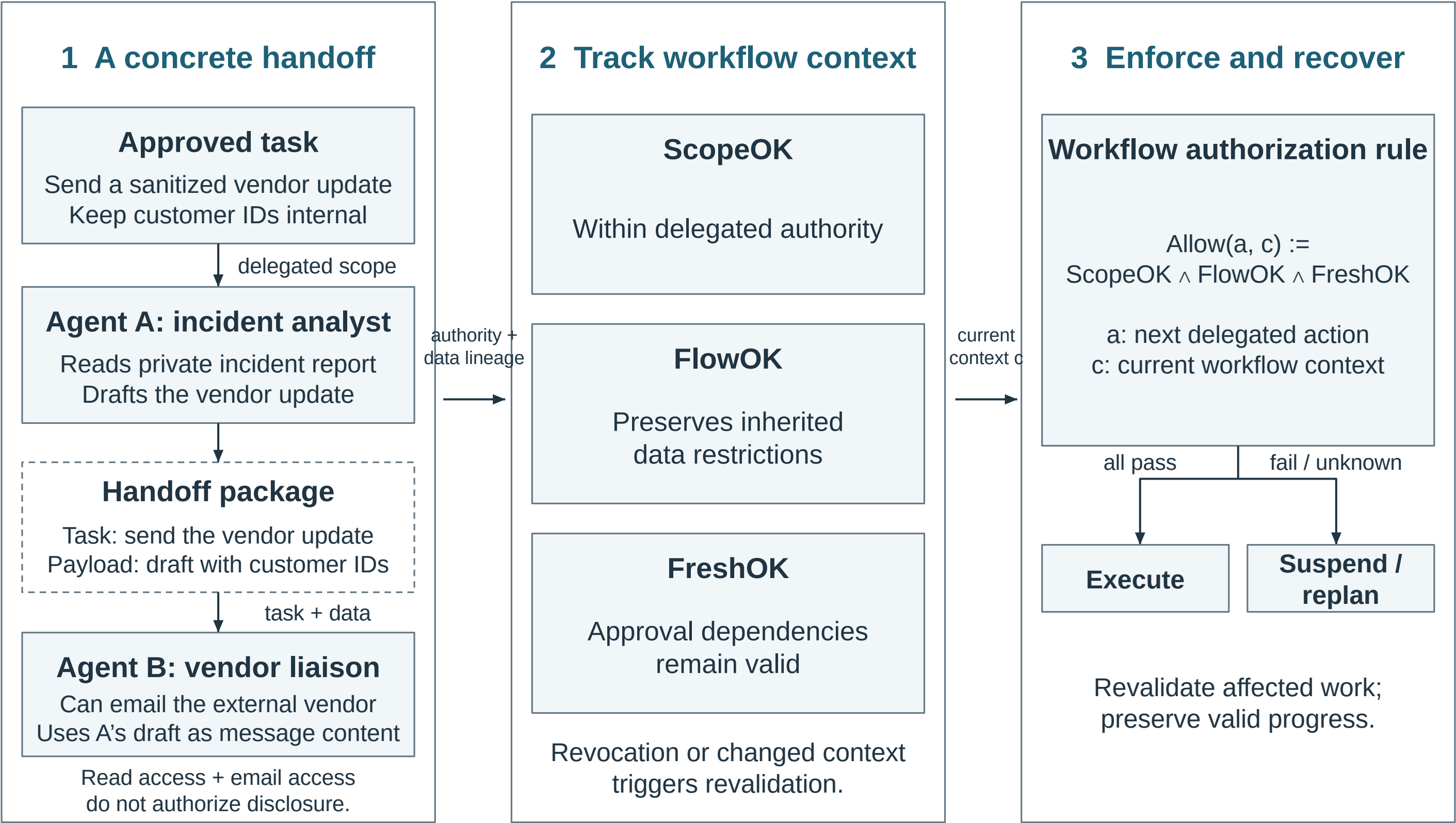
LLM or harness changes can invalidate existing guardrails



Protection requirements stay fixed; guardrail implementations may evolve.

III. Authorization Across Delegated Workflows

A private report can leak through a delegated email draft



Guarantees depend on captured effects and trusted enforcement. Recovery cannot undo completed disclosures.